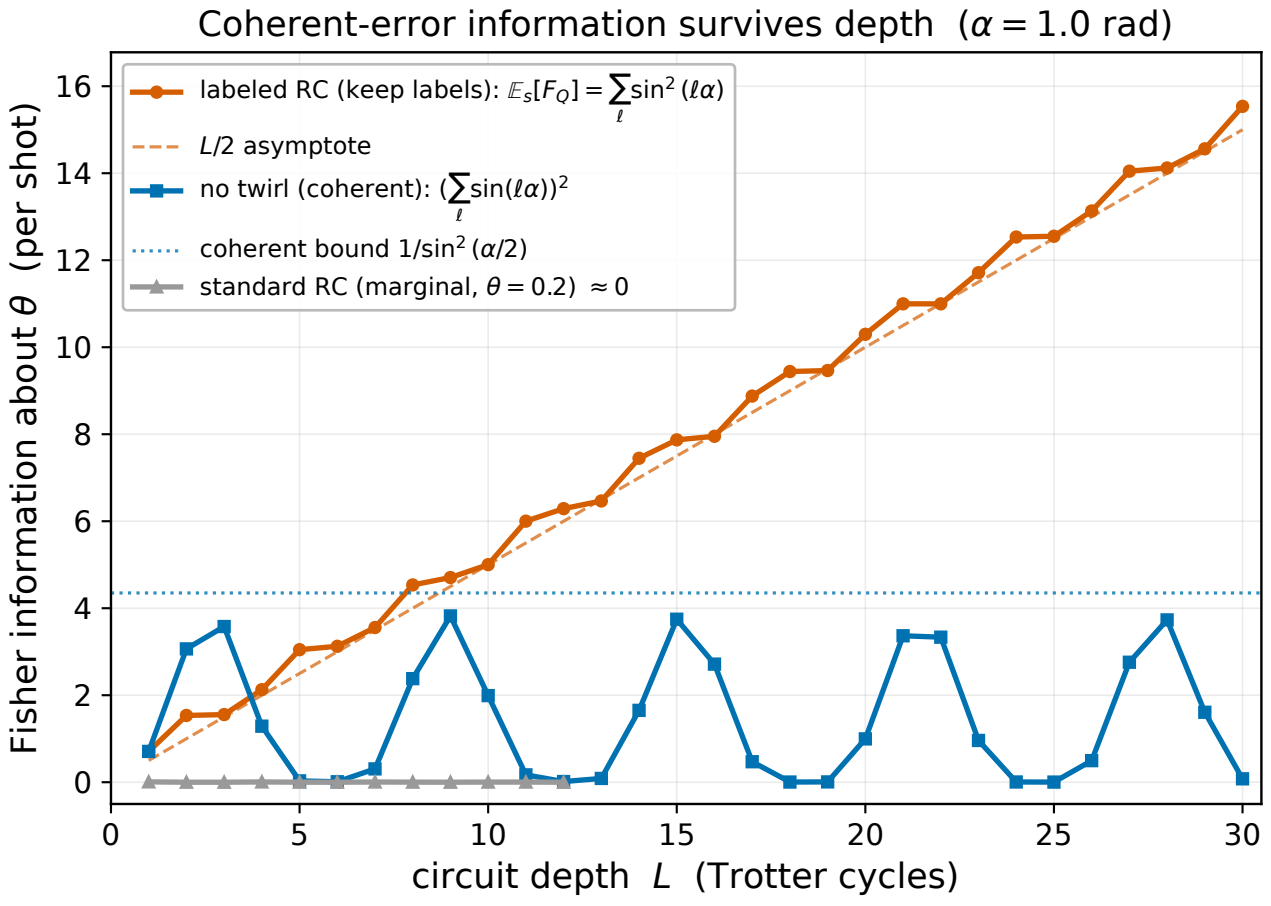
Coherent-error information survives depth ($\alpha = 1.0$ rad)
labeled RC (keep labels): $\mathbb{E}_s[F_Q] = \sum_\ell \sin^2(\ell\alpha)$
$L/2$ asymptote
no twirl (coherent): $(\sum_\ell \sin(\ell\alpha))^2$
coherent bound $1/\sin^2(\alpha/2)$
standard RC (marginal, $\theta = 0.2$) ≈ 0
Fisher information about θ (per shot)
circuit depth L (Trotter cycles)
0
2
4
6
8
10
12
14
16
0
5
10
15
20
25
30